Fiscal Year:	FY 2014	**Task Last Updated:**	FY 12/27/2013
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Project Title:	Two-Hit Model of Alzheimer's Disease		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:	**TechPort:**		No
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Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44106-3860	**Congressional District:**	11
Comments:			
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Flight Program:			
Flight Assignment:			
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COI Name (Institution):	Casadesus Smith, Gemma (Case Western Reserve University) Lee, Hyoung-gon (Case Western Reserve University) Wang, Xinglong (Case Western Reserve University)		
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Task Description:	Future missions in space will involve long duration travel beyond the magnetic field of the Earth. During these long-duration spaceflights, astronauts will be exposed to types and doses of radiation (cosmic rays) that are not experienced in low Earth orbit. The hazards associated with the space radiation will likely impact various organs and systems and there is a persistent effect of exposure to HZE particles lasting beyond the time of exposure. Space radiation risk to the central nervous system (CNS) is a major concern to NASA. It is known that space-like radiation induces neuronal and behavioral deficits, including hippocampus-dependent cognitive impairments which are also features in Alzheimer's Disease (AD) patients. Indeed, there are similarities between the effects of radiation and AD including cognitive deficits and neurochemical abnormalities which calls for a detailed investigation on the potential risk for AD of space radiation. In this proposal, we aim at exploring the potential mechanisms by which exposure to HZE particles and proton leads to the development of long-term degenerative processes in neuronal function and cognitive performance related to AD and the relationship of these effects to radiation parameters (particle, dose, and LET) and individual characteristics (age and gender) in AD model based on a novel Two Hit Hypothesis of AD pathogenesis. Two specific aims are proposed: Aim 1: To determine whether space radiation induces AD-relevant pathological changes in fully differentiated primary hippocampal neurons from CRN8 APP Tg mouse and 3xTg AD mouse models. Aim 2: To determine whether space radiation induces neurodegeneration and exacerbate AD-relevant pathological/cognitive/behavioral changes in 3xTg AD mice. These proposed studies will specifically address many of the critical issues NASA has identified.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	Detrimental effects of space irradiation on the CNS have been known for some time and emerging evidence supports a potential risk for AD as a valid concern for long duration space exploration such as lunar mission or a Mars mission. Based on our study on mitotic and oxidative signaling and the recent development in AD field, we proposed a novel Two Hit Hypothesis stating that both oxidative stress and aberrant mitotic stimuli can independently initiate, yet both are necessary to propagate disease pathogenesis and progression. Our hypothesis predicts that after certain threshold of exposure, space irradiation will cause increased oxidative stress in genetically predisposed individuals who are under mitotic stress and induce full blown AD. Therefore, our two hit hypothesis not only calls for a detailed examination of AD risk during space radiation exposure, if confirmed, but also provide the potential mechanisms and working model to estimate risk for space crew to develop AD.		

Task Progress:	This is the third and last year of our project. We have participated in the spring and fall run in 2013 and irradiated additional CRND8 APP transgenic mice to supplement the loss of mice due to shipment problems during the fall run in 2012. We performed behavioral tests and collected and analyzed all the brain samples as planned. We also repeated experiments to evaluate the effectiveness of antioxidant and/or cell cycle inhibitor in rescuing neuronal death after space irradiation in the primary culture and explore the mitochondrial dysfunction as the upstream event of oxidative stress during irradiation. For aim 1, we repeated the inhibition experiments [cell cycle inhibitors (i.e., olomoucine, roscovitine, tenovin-1 and Flavopiridol) and/or antioxidants (i.e., N-acetyl-cysteine (NAC); vitamine E and lipoic acid)] to determine the causal relationship between oxidative stress and cell cycle alterations and the subsequent cell death and obtained similar results as we reported in the progress report last year. To explore the upstream events that may cause increased oxidative stress after irradiation, we determined various mitochondrial parameters with a focus on mitochondrial dynamics. Primary hippocampal neurons (DIV14) from WT or CRND8 Tg mice were exposed to 0.5 Gy 56Fe (600Mev/u) and fixed 24 hrs later. Mitochondrial morphology was studied after immunostaining by COX-IV antibody. Interestingly, WT neurons exposed to 0.5Gy 56Fe demonstrated fragmented mitochondria, very similar to that in CRND8 neurons. Further mitochondrial fragmentation and mitochondrial aggregates were often observed in CRND8 neurons exposed to 56Fe. To have a better appreciation of changes in mitochondrial dynamics, live imaging study was also performed. Primary hippocampal neurons were transfected with MitoDsRed to label mitochondria before irradiation. 24hr after irradiation, mitochondria in living neurons (DIV14) were imaged for up to 15 min under fluorescent microscope for time-lapse recordings. Consistent with the studies in fixed cells, CRND8 neurons demonstrated more fragmented mitochondria than WT neurons. Exposure to 0.5Gy 56Fe caused mitochondrial fragmentation in WT neurons which was further exacerbated in CRND8 neurons. Importantly, live imaging revealed additional information: radiation caused significantly reduced mitochondrial density in the axons of WT neurons. This is likely due to the effects on mitochondrial transport since radiation specifically inhibits the retrospective axonal transport of mitochondria in both WT and CRND8 neurons. Therefore, live imaging study suggests that radiation can negatively impact both mitochondrial morphology, distribution and transport similar to that happened in AD. For aim 2, we finished all the pathological and behavioral analysis as we planned. Two separate tests were employed to assess cognitive/behavioral deficits. For hippocampus-dependent memory, we used contexture fear conditioning tests at 1 month post irradiation. We found an overall significant difference in the freezing behavior as measured by one-way ANOVA in the male group but not the female group. Post-hoc analysis revealed a significant decrease in the freezing behavior in male CRND8 mice exposed to either 0.5Gy 56Fe, 2 Gy 56Fe or 2Gy 28Si compared to non-irradiated CRND8 male mice. Male CRND8 mice exposed to 0.5 Gy 28Si also showed a trend towards decreased freezing although not reaching significance. The mice were also subject to novel object cognition test at 2 months post irradiation. We also found an overall significant difference and post-hoc analysis revealed a significant decrease in exploratory time spent with the novel objects in mice exposed to either 0.5Gy 56Fe, 2 Gy 56Fe or 0.5Gy 28Si compared to non-irradiated CRND8 mice. CRND8 mice exposed to 2 Gy 28Si also showed a trend towards decreased exploratory time spent with novel object although not reaching significance. Unlike the fear conditioning test, no difference in gender was noted since both male and female demonstrated loss of novel object recognition after radiation. Pathological analysis was performed on all the animals collected which generally confirmed our observation during the pilot study with significance. For example, quantitative analysis revealed that CRND8 mice exposed to 0.5Gy or 2 Gy 56Fe (i.e., 3 month post-irradiation in mice irradiated at 6 months old) demonstrated significantly increased plaque load compared to non-irradiated control CRND8 mice. There is a trend towards increased plaque load in mice exposed to 0.5 Gy or 2 Gy 28Si (i.e., 3 month post-irradiation in mice irradiated at 6 months old). Interestingly, this increased plaque load trend became significant if mice exposed to either 0.5 Gy or 2 Gy 28Si were sacrificed 3 months later (i.e., 6 month post-irradiation in mice irradiated at 6 months old), suggesting that 28Si has more significant effect later during the progression of the pathology. It is believed that oxidative stress plays a critical role in the pathogenesis of AD. To explore the mechanisms underlying exacerbated pathology and cognitive/behavioral deficits-induced by radiation, we sought to determine whether there is enhanced oxidative stress. We examined 9 months old mouse brains immunocytochemically with antibody against 3-nitrotyrosine (3-NT) as a marker for oxidative damage to proteins. Consistent with the prior reports of the presence of oxidative stress in APP transgenic mice, there is elevated basal 3-NT immunoreactivity in the brain of non-irradiated CRND8 mice. Furthermore, we found significantly increased immunoreactivity in the hippocampus from mice exposed to 0.5Gy 56Fe (3 months post-irradiation), suggesting that HZE particles caused enhanced oxidative stress in CRND8 mice. We also examined mouse brain immunocytochemically with antibody recognizing oxidized nucleosides, 8-OHdG, which are markers of oxidative damage to DNA and found that exposure to 0.5 Gy 56Fe caused significantly increased immunoreactivity in the pyramidal neurons of hippocampus, again confirming the enhanced oxidative stress in irradiated mouse brain. The most striking feature was the prominent immunolabeling of 8OHdG in the neuronal cytoplasmic distribution extending from the cell bodies to processes. Intranuclear immunoreaction with 8-OHdG in exposed brain was less intense than that found for the cytoplasmic immunoreactions. To exclude the possibility that the cytoplasmic staining is due to cross-reactivity with 8OHG, markers of oxidative damage to RNA, we pretreated sections with DNases which essentially abolished 8OHdG immunoreactivity while RNase pretreatment had little effects, thus suggesting that the increased immunoreactivity reflects increased oxidative damage to DNA in the cytoplasm, which are likely mtDNA. Guided by these novel findings, at the last stage of this funding period, we began to study mitochondrial dysfunction. To determine whether mitochondrial dysfunction is involved in radiation-induced exacerbation of AD phenotype, mitochondrial respiration was determined in freshly isolated synaptic mitochondria from hippocampus of 12 month old CRND8 mice that were irradiated 6 months ago using a Seahorse XF24 extracellular flux analyzer by following the technical details specifically developed for oxygen consumption rate (OCR) analysis of isolated mitochondrial preparations. Our data suggest impairments in the maximal respiratory capacity and reserve respiratory capacity (difference between basal and maximal respiratory capacity, critical for neuronal bioenergetics under stress) in the synaptic mitochondria from hippocampus of CRND8 mice exposed to 0.5 Gy 56Fe six months ago. Our data demonstrated long-lasting detrimental effects of irradiation (i.e., 6 months post-irradiation) of low levels of HZE particles (i.e., 0.5Gy 56Fe) on mitochondrial function which likely contributes to increased oxidative stress and exacerbated AD-phenotype in CRND8 mice exposed to radiation. In fact, irradiation-induced abnormal mitochondrial changes can occur very rapidly. In an experiment with more rapid setting (3 or 7 days post-irradiation), we found that 6 months old CRND8 mice exposed to either 0.5Gy or 2 Gy 56Fe demonstrated significantly decreased expression of critical components in the electron transport complexes as early as 3 days post-irradiation, which is likely the molecular basis of impaired respiration. These data suggest that mitochondrial dysfunction could also contribute to acute effects induced by exposure to HZE particles.
Bibliography Type:	Description: (Last Updated: 04/11/2018)
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